UNIT: I

IT Act and Encryption

❖ What is IT Act:

The Information Technology Act, 2000, also known as the IT Act, was passed by the Indian Parliament on 17th October 2000. It provides the legal framework for electronic transactions, digital communication, and cybercrimes in India. This

3law was designed to promote e-governance, regulate online transactions, and protect against cybercrimes.

The Information Technology Act, 2000 is India's primary law governing digital activities. It was enacted to:

- Provide legal recognition to electronic records and digital signatures.
- Regulate cybercrime, hacking, and data protection.
- Enable secure electronic transactions and e-governance.

The main goal of the IT Act is to promote secure digital transactions, make cybercrimes easier to handle, and ensure e-governance, and individual use the internet safely.

The Act has 94 sections and is divides into 13 chapters. The last few sections focus on changes to India's older law, the Indian Penal Code, 1860.

❖ Object and Scope of the IT Act

Object of the IT Act :

- **Legal Recognition of Electronic Transactions:**
  To give legal validity to electronic records and digital signatures, making them equivalent to paper-based documents and handwritten signatures.

- **Facilitate E-Governance:**
  To enable government departments to accept and process documents electronically, improving efficiency and accessibility.
- **Promote E-Commerce:**
  To support secure online business transactions and digital communication between companies, banks, and consumers.
- **Establish Regulatory Bodies:**
  To create authorities like the Controller of Certifying Authorities (CCA) for managing digital signatures and certificates.
- **Enable Electronic Storage of Data:**
  To allow banks and institutions to maintain records in digital formats legally.
- **Prevent and punish cyber-crimes:**
  It defines crimes like hacking, identity theft, and online fraud, and sets penalties for them.
- **Protect user privacy and data:**
  It ensures that personal data is handled responsibly and securely.

Scope of the IT Act:

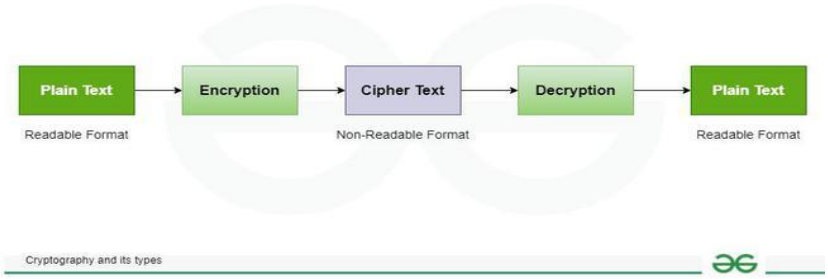
- **Applies to All of India:**
  It's valid across the country.
- **Offenses committed outside India:**
  If someone outside India commits a cybercrime involving a computer or network in India, this law still applies.
- **All digital activities:**
  Including emails, digital contracts, websites, online banking, social media, mobile apps, and e-commerce platforms.
- **Electronic transactions and communication:**
  It cover how people exchange information and money online.
- **Exclusions:**
  It does *not* apply to things like wills, trusts, power of attorney, or sale of immovable property unless notified by the government.

❖ What is Cryptography?

Cryptography is the science and practice of secure communication by using mathematical algorithms to transform information into an unreadable format, known as ciphertext, that only authorized individuals can decipher back into plain text using a secret key. Cryptography is fundamental to modern cybersecurity, securing everything from online banking and e-commerce to digital currencies and secure messaging.

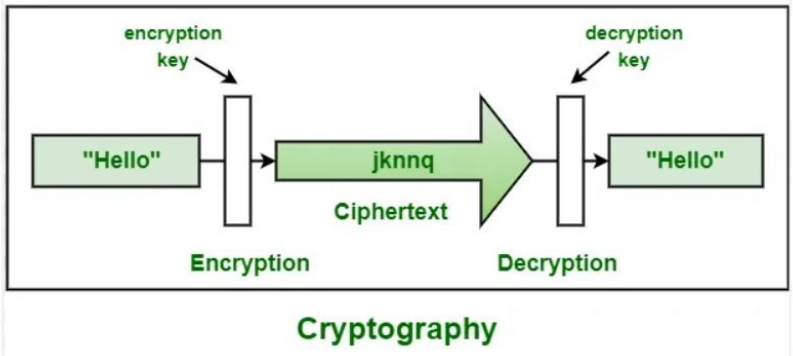
Key Concepts -

- **Plain text:** The original, readable form of information.
- **Cipher text:** The scrambled, unreadable version of the information.
- **Encryption:** The process of converting plain text into cipher text.
- **Decryption:** The process of converting cipher text back into plain text.
- **Algorithm:** A set of rules or steps used to perform encryption and decryption.
- **Key:** A secret piece of information, usually a string of data, that is used with an algorithm to encrypt and decrypt messages.



Working of Cryptography

As we all know that cryptography technique is use to convert plain text into ciphertext. This technique is done by cryptographic key. Basically cryptographic key is a string of characters which is used to encrypts the data and decrypt the data.

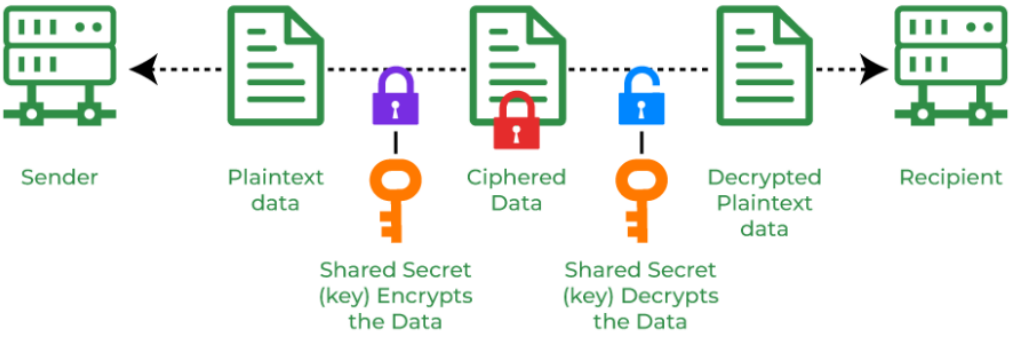


Here,

"Hello" is a plaintext and convert into ciphertext "jknnq" with the help of cryptographic key and then decrypt into "Hello".

❖ Symmetric Key Cryptography

Symmetric Key Cryptography is an encryption system where the sender and receiver of a message use a single common key to encrypt and decrypt messages. Symmetric Key cryptography is faster and simpler but the problem is that the sender and receiver have to somehow exchange keys securely. The most popular symmetric key cryptography systems are Data Encryption Systems (DES) and Advanced Encryption Systems (AES).

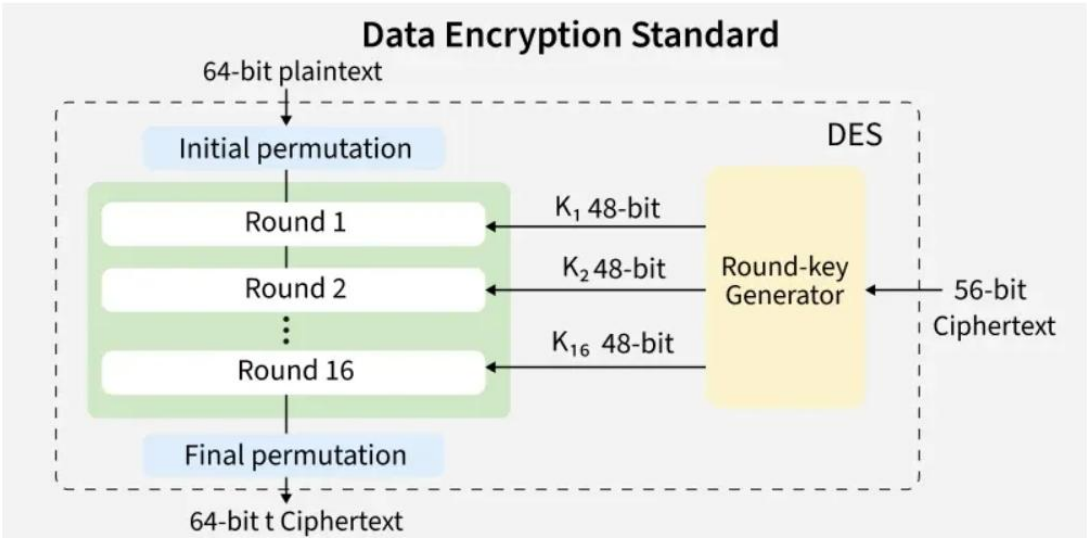


Symmetric Key Cryptography

Example:

1. Data Encryption Standard (DES)

DES (Data encryption standard) is an older encryption algorithm that is used to convert 64-bit plaintext data into 48-bit encrypted ciphertext. It uses symmetric keys (which means same key for encryption and decryption). It is kind of old by today's standard but can be used as a basic building block for learning newer encryption algorithms.



2. AES (Advanced Encryption Standard):

AES s a popular encryption algorithm which uses the same key for encryption and decryption It is a symmetric block cipher algorithm with block size of 128 bits, 192 bits or 256 bits. AES algorithm is widely regarded as the replacement of DES (Data encryption standard) algorithm.

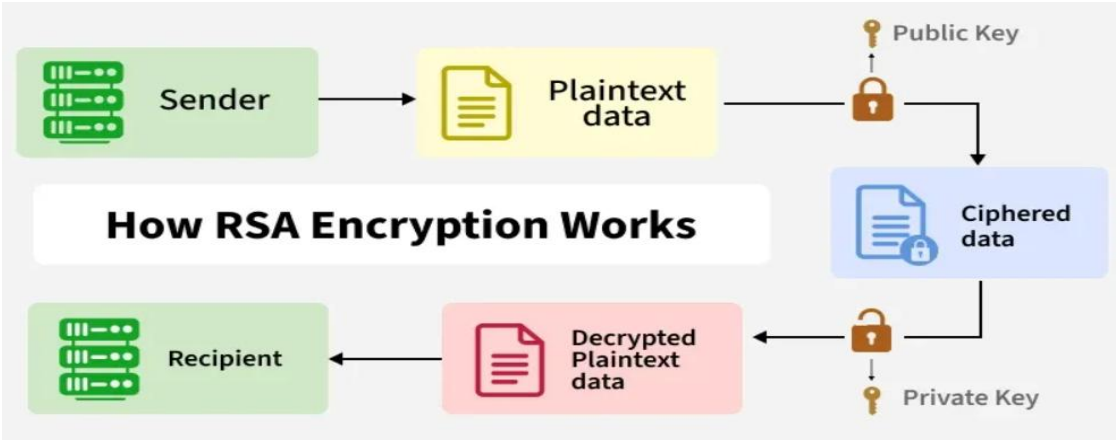
❖ Asymmetric Key Cryptography

In Asymmetric Key Cryptography a pair of keys is used to encrypt and decrypt information. A sender's public key is used for encryption and a receiver's private key is used for decryption. Public keys and Private keys are different. Even if the public key is known by everyone the intended receiver can only decode it because he holds his private key. The most popular asymmetric key cryptography algorithm is the RSA algorithm.

Examples:

RSA (Rivest-Shamir-Adleman):

RSA is an basic asymmetric cryptographic algorithm which uses two different keys for encryption. The RSA algorithm works on a block cipher concept that converts plain text into cipher text and vice versa.



❖ RSA Algorithm in Cryptography

RSA(Rivest-Shamir-Adleman) Algorithm is an asymmetric or public-key cryptography algorithm which means it works on two different keys: Public Key and Private Key. The Public Key is used for encryption and is known to everyone, while the Private Key is used for decryption and must be kept secret by the receiver. RSA Algorithm is named after Ron Rivest, Adi Shamir and Leonard Adleman, who published the algorithm in 1977.

Example of Asymmetric Cryptography:

If Person A wants to send a message securely to Person B:

- Person A encrypts the message using Person B's Public Key.
- Person B decrypts the message using their Private Key.

❖ RSA Algorithm

RSA Algorithm is based on factorization of large number and modular arithmetic for encrypting and decrypting data. It consists of three main stages:

1. ***Key Generation**: Creating Public and Private Keys*
2. ***Encryption**: Sender encrypts the data using Public Key to get cipher text.*
3. ***Decryption**: Decrypting the cipher text using Private Key to get the original data.*

<u>1. Key Generation:</u>

- Choose two large prime numbers, say **p** and **q**. These prime numbers should be kept secret.
- Calculate the product of primes, **n = p * q**. This product is part of the public as well as the private key.

Example:

Choose two Prime Numbers: p=3, q=11

*Calculate: n=p*q = 3*11= 33*

*(n) = (p-1) (q-1) = 2*10=20*

Let's choose e = 7 (a common choice or co-prime with 20).

Calculate d as the multiplicative inverse of e(7), so d =3

Finally, the ***Public Key = (n, e)*** *= (33, 7) and the* ***Private Key = (n, d)*** *= (20, 3).*

2. Encryption:

To encrypt a message **M**, it is first converted to numerical representation using ASCII and other encoding schemes. Now, use the public key (n, e) to encrypt the message and get the cipher text using the formula:

C = Me mod n*, where C is the Cipher text and e and n are parts of public key.*

3. Decryption:

To decrypt the cipher text **C**, use the private key (n, d) and get the original data using the formula:

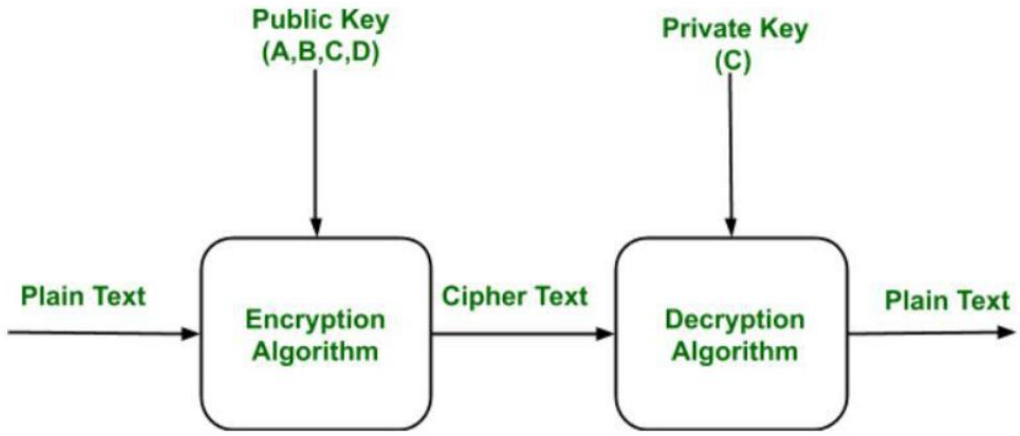
M = Cd mod n, *where M is the message and d and n are parts of private key.*

❖ Public Key Encryption

Public key encryption, also known as asymmetric encryption, uses a pair of mathematically linked keys – a public key and a private key – for secure communication. The public key is freely distributed and used to encrypt data, while the corresponding private key, which must be kept secret by the owner, is used to decrypt it.

Components of Public Key Encryption-

- **Plain Text:** This is the message which is readable or understandable. This message is given to the Encryption algorithm as an input.
- **Cipher Text:** The cipher text is produced as an output of Encryption algorithm. We cannot simply understand this message.
- **Encryption Algorithm:** The encryption algorithm is used to convert plain text into cipher text.
- **Decryption Algorithm:** It accepts the cipher text as input and the matching key (Private Key or Public key) and produces the original plain text.
- **Public and Private Key:** One key either Private key (Secret key) or Public Key (known to everyone) is used for encryption and other is used for decryption.

How it works:

1. **Encryption:** To send a secure message, you use the recipient's public key to encrypt the message.
2. **Transmission:** The encrypted message is then sent over a network.
3. **Decryption:** Upon receiving the message. The recipient uses their private key to decrypt the message and recover the original plaintext